

Amr Abed

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Experienced software engineer and leader specializing in machine learning solutions. Proven track record in building and deploying large-scale machine learning models on cloud platforms. Skilled in designing end-to-end machine learning systems using TensorFlow, Keras, and Scikit-Learn. Expertise in cloud architecture, backend development, and API design. AWS Certified Machine Learning Specialist and Google Cloud Certified. Capable of leading engineering teams to deliver innovative AI products.

SELECT EXPERIENCE

SOPHI | HANDS-ON ENGINEERING MANAGER, MACHINE LEARNING
Nov 2022 - Present | Toronto, ON | Remote

- Led the development of Sophi Paywall products, delivering ML solutions to optimize content monetization
- Automated deployment of ML models using MLFlow and Amazon EKS
- Reduced operational costs through consolidating systems into a multi-tenant architecture

SOPHI | LEAD SOFTWARE ENGINEER, MACHINE LEARNING
Apr 2022 - Nov 2022 | Toronto, ON | Remote

- Led Print Laydown automation, optimizing workflows with AWS-based backend services using Python
- Managed a team of 3 engineers to architect and implement cloud solutions on AWS
- Collaborated with DevOps to optimize cloud services for scalability

SOPHI | SENIOR SOFTWARE ENGINEER, MACHINE LEARNING
Jan 2021 - Apr 2022 | Toronto, ON | Remote

- Deployed TensorFlow deep learning models in AWS for real-time decision-making processes
- Developed and maintained scalable backend systems for AI-driven decision-making using Python
- Built APIs to enable integration of ML services with user-facing and admin applications

ROBOGARDEN | LEARNING CONTENT TEAM LEAD
Dec 2018 - Aug 2019 | Calgary, AB | Remote

- Created a Python ML course for an experiential learning program of a top Canadian university
- Led a team of 6 engineers to deliver content for 6 different courses

VIRGINIA TECH | GRADUATE RESEARCH ASSISTANT
Jan 2015 - Jul 2017 | Blacksburg, VA

- Collaborated with engineers from Northrop Grumman, Fujitsu, and VT Urban Living Lab to complete 3 industry-funded research projects
- Built the Strace-Docker 🔗 and RHIDS 🔗 tools using Java, Docker, and shell scripting to monitor Docker containers in a cloud environment

GOOGLE | SOFTWARE ENGINEERING INTERN
May 2014 - Aug 2014 | Mountain View, CA

- Implemented an Android application powered by Google Visual Search and Google Knowledge Graph APIs
- Added features and fixed bugs for two other Android apps using Google Fit API

SKILLS

PROGRAMMING

Python • Java • C/C++ • Kotlin • Swift

AI & ML

TensorFlow • Keras • Scikit-Learn

MLOPS

MLFlow • Docker • Kubernetes

CLOUD

AWS • Google Cloud • Firebase

EDUCATION

VIRGINIA TECH

PH.D. COMPUTER ENGINEERING
2014 - 2017

VIRGINIA TECH

M.S. COMPUTER ENGINEERING
2010 - 2013

AIN-SHAMS UNIVERSITY

B.S. COMPUTER ENGINEERING
2000 - 2005

CERTIFICATIONS

AWS CERTIFIED

GEN AI DEVELOPER - PROFESSIONAL
March 2026

MACHINE LEARNING - SPECIALITY
June 2024

SOLUTIONS ARCHITECT - ASSOCIATE
October 2020

DEEPLARNING.AI

TENSORFLOW DEVELOPER
September 2020

SCRUM ALLIANCE

CERTIFIED SCRUM MASTER
CERTIFIED SCRUM PRODUCT OWNER
June 2020

CITIZENSHIP

Canadian • Egyptian

LANGUAGES

Arabic • English • French

MORE EXPERIENCE

MINIA UNIVERSITY | ASSISTANT PROFESSOR OF COMPUTER ENGINEERING

Aug 2017 - Jan 2020 | Minia, Egypt

- Supervised a Machine-Learning-based Mobile Cloud Computing graduation project
- Created 3 undergraduate courses, including Cloud Computing and Database courses

MENTOR (now Siemens EDA) | SOFTWARE DEVELOPMENT ENGINEER

Nov 2007 - Nov 2008 | Cairo, Egypt

- Studied and analyzed the C-based version of the Spectre2Eldo syntax converter
- Designed and implemented the OO-based Spectre2Eldo syntax converter using STL

MINIA UNIVERSITY | TEACHING ASSISTANT

Mar 2007 - Aug 2010 | Minia, Egypt

- Taught 8 different undergraduate courses to 4 different classes of an average of 25 students each

SHOROUK ACADEMY | TEACHING ASSISTANT

Jan 2006 - Sep 2007 | Cairo, Egypt

- Taught 6 different undergraduate courses to 9 different classes of an average of 40 students each

SELECT PROJECTS

PAYWALL ENGINES | SOPHI

Jan 2021 - Present

ML-powered Paywall engines for online publications

PYTHON FOR MACHINE LEARNING COURSE | ROBOGARDEN

Dec 2018 - Aug 2019

A course to introduce learners from different backgrounds to the concepts of machine learning using Python

INTRODUCTION TO PROGRAMMING COURSE | MINIA UNIVERSITY

Sep 2018 - Dec 2018

A course to introduce first-year Computer Engineering students to programming using C++

RESILIENT HOST-BASED INTRUSION DETECTION SYSTEM | VIRGINIA TECH

Jan 2015 - July 2017

Ph.D. dissertation project - An LSTM-based Deep Learning model using TensorFlow for anomaly-based Intrusion Detection System for Docker Containers running in a cloud environment

PUBLICATIONS

Amr Abed. **Securing Cloud Containers through Intrusion Detection and Remediation**. Ph.D. dissertation, Virginia Tech, 2017.

Amr Abed, Mohamed Azab, Charles Clancy, and Mona Kashkoush. **Resilient Intrusion Detection System for Cloud Containers**. *International Journal of Communication Networks and Distributed Systems*, 2018.

Amr Abed, Charles Clancy, and David Levy. **Applying Bag of System Calls for Anomalous Behavior Detection of Applications in Linux Containers**. In *IEEE GlobeCom 2015 Workshop on Cloud Computing Systems, Networks, and Applications (CCSNA)*, 2015.

Amr Abed, Charles Clancy, and David Levy. **Intrusion Detection System for Applications Using Linux Containers**. In *Security and Trust Management*, volume 9331 of *Lecture Notes in Computer Science (LNCS)*, pages 123–135, 2015.

Mohamed Azab, Bassem Mokhtar, Amr Abed, and Mohamed Eltoweissy. **Smart Moving Target Defense for Linux Container Resiliency**. In *2016 IEEE 2nd International Conference on Collaboration and Internet Computing (CIC)*, pages 122–130, 2016. Invited paper.

Mohamed Azab, Bassem Mokhtar, Amr Abed, and Mohamed Eltoweissy. **Toward Smart Moving Target Defense for Linux Container Resiliency**. In *IEEE 41st Conference on Local Computer Networks (LCN)*, 2016.

Mona Kashkoush, Mohamed Azab, Gamal Attiya, and Amr Abed. **Online Smart Disguise: Real-time Diversification Evading Coresidency Based Cloud Attacks**. *Cluster Computing*, 2018.